A Machine Learning Framework for Pitting Corrosion Detection in Offshore Infrastructure Using Fused RGB, Thermal, and LiDAR Imagery

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***Abstract*—Pitting corrosion is one of the most hazardous and economically costly forms of material degradation affecting offshore oil and gas infrastructure, including fixed platforms, subsea pipelines, and floating production systems. Conventional automated inspection approaches rely predominantly on single-modality sensing—visible-spectrum (RGB) imagery, thermal infrared, or structured-light scanning in isolation—which constrains detection reliability under the variable illumination, surface fouling, and harsh environmental conditions characteristic of offshore deployments. This paper proposes a multimodal machine learning (ML) framework that fuses RGB, thermal infrared, and LiDAR point-cloud data for the automated detection and localisation of pitting corrosion on offshore structures. By exploiting the complementary strengths of each modality—colour and texture discrimination from RGB, subsurface thermal anomaly signatures from infrared, and high-fidelity geometric surface deviation from LiDAR—the proposed hybrid feature-level fusion architecture aims to achieve detection accuracy and environmental robustness that no single-modality approach can deliver alone. The framework is to be evaluated against single- and dual-modality baselines using standardised metrics including intersection over union (IoU), precision, recall, and F1-score. This research addresses a documented gap in the literature: no existing study has unified all three sensor modalities specifically for pitting corrosion detection in offshore environments. Expected contributions include a novel trimodal fusion framework, a benchmark multimodal corrosion dataset, and quantified accuracy improvements, with direct implications for structural safety and inspection efficiency in the offshore energy sector.**

***Index Terms*—pitting corrosion, multimodal fusion, LiDAR, thermal infrared imaging, convolutional neural networks, offshore structural health monitoring**

I. Introduction

Offshore oil and gas infrastructure operates in one of the most corrosive environments on Earth. Sustained exposure to marine aerosols, saltwater immersion, cyclic thermal loading, and biological fouling accelerates corrosion processes that, if undetected, can precipitate catastrophic structural failure, environmental contamination, and substantial economic loss [1]. Among the various corrosion morphologies, pitting corrosion is particularly insidious: its localised, subsurface-nucleating character means that significant material loss can develop with minimal macroscopic surface signature until the corrosion penetrates through the wall of a pipe or plate [2].

Current offshore inspection regimes combine periodic manual visual inspection with single-sensor non-destructive testing (NDT) techniques. Whilst RGB camera systems have gained traction for automated surface inspection due to their low deployment cost, they degrade markedly under poor illumination, surface contamination, and adverse weather [3]. Thermal infrared (IR) imaging can reveal subsurface thermal anomalies indicative of material thinning but is limited in spatial resolution and prone to false positives from environmentally induced thermal gradients [4]. LiDAR sensors provide dense three-dimensional point clouds encoding surface topology with sub-millimetre precision, enabling pit detection as geometric depressions, yet are susceptible to noise from water films, maritime aerosols, and specular reflectance on metallic substrates [5].

Multimodal sensor fusion—exploiting complementary sensing physics to improve system-level performance—is well established in autonomous driving, aerial remote sensing, and medical imaging. However, its systematic application to pitting corrosion detection on offshore structures remains largely unexplored. This gap represents both a scientific opportunity and a practical imperative given the industry's need for more reliable, autonomous inspection solutions.

This research proposes a multimodal ML framework fusing RGB, thermal, and LiDAR data for pitting corrosion detection on offshore infrastructure. The specific research objectives are:

1) To design a hybrid feature-level fusion architecture integrating RGB, thermal infrared, and LiDAR modalities for pitting corrosion detection on offshore structural steel.
2) To curate or synthesise a co-registered multimodal corrosion dataset representative of offshore environmental conditions, including surface fouling, variable illumination, and geometric irregularity.
3) To evaluate the proposed framework against single-modality (RGB-only, IR-only, LiDAR-only) and dual-modality baselines using standardised pixel-level and detection-level metrics.
4) To assess the robustness and generalisability of the framework under controlled simulations of realistic offshore degradation scenarios.

The remainder of this paper is organised as follows: Section II reviews related work. Section III presents the proposed

methodology. Section IV discusses expected outcomes. Section V outlines the research timeline. Section VI concludes the paper.

II. Related Work

A. RGB-Based Corrosion Detection

Visible-spectrum imaging is the most widely studied modality for automated corrosion detection. Early work employed handcrafted colour and texture features [6], whilst more recent studies have leveraged deep CNNs, including YOLO-based detectors and encoder-decoder segmentation networks, achieving F1-scores exceeding 0.90 on curated laboratory benchmarks [7]. Transfer learning from large-scale visual datasets has partially improved generalisation, but RGB-based models remain vulnerable to illumination variation, specular reflectance, and surface contamination—all of which are endemic in offshore environments [8]. Critically, RGB imaging provides no information about subsurface material loss or surface topography, limiting its diagnostic capability for pitting corrosion.

B. Thermal Infrared-Based Detection

Infrared thermography is an established NDT technique in which subsurface defects—including corrosion-induced wall thinning—are revealed as thermal anomalies under passive or active thermal excitation [9]. ML methods applied to thermal imagery, including SVMs, CNNs, and U-Net segmentation architectures, have demonstrated feasibility for corrosion characterisation [10]. Thermal imaging offers reduced sensitivity to ambient lighting relative to RGB, enabling inspection in low-visibility or night-time conditions. However, its spatial resolution is generally inferior to optical cameras, and discriminating between corrosion-induced and environmentally induced thermal gradients remains an open problem in offshore deployments [11].

C. LiDAR-Based Structural Inspection

LiDAR sensors generate dense 3D point clouds with high geometric fidelity, enabling pitting to be detected as localised surface depressions [12]. Deep learning approaches for point cloud processing—including PointNet, PointNet++, and sparse 3D CNNs—have been applied to structural inspection with promising results [13]. In marine and offshore contexts, LiDAR has been used for hull inspection and pipeline monitoring, demonstrating capability to resolve geometric features invisible to optical sensors [14]. Known challenges include sensitivity to water film, maritime aerosol scatter, and specular reflectance from metallic surfaces, all of which introduce noise in offshore point cloud acquisitions.

D. Multimodal Fusion Approaches and the Research Gap

Multimodal fusion has consistently outperformed single-modality baselines in domains including autonomous driving [15], aerial remote sensing [16], and medical image analysis [17]. Within structural inspection, limited studies have fused RGB with thermal data for surface defect detection [18] or RGB with LiDAR for damage assessment [19]. To the best of the author's knowledge, no existing study has proposed a unified framework fusing all three modalities—RGB, thermal infrared, and LiDAR—specifically for pitting corrosion detection on offshore infrastructure. This gap motivates the present research.

III. Proposed Methodology

A. Research Design

This study follows a quantitative experimental design. The primary instrument is the proposed multimodal ML framework, evaluated through controlled experiments on a purpose-assembled dataset. Ablation studies isolating each modality's contribution, combined with baseline comparisons, provide the empirical foundation for evaluating the fusion approach.

B. Data Acquisition and Preparation

A multimodal dataset will be compiled from three complementary sources:

1) **Public benchmarks:** Existing corrosion and surface defect datasets (e.g., CODEBRIM [20]; NEU surface defect database [21]) will be used for pre-training and transfer learning initialisation. Where paired thermal or 3D scan data are available, they will be incorporated.
2) **Synthetic data generation:** Pitting textures, thermal anomaly maps, and point-cloud surface deformations will be synthesised using physically-based rendering pipelines and corrosion simulation models. Synthetic samples extend training coverage and allow controlled variation of pit depth, density, and environmental conditions.
3) **Laboratory specimen imaging:** Corroded steel coupons representative of offshore structural grades will be imaged using an RGB camera, an uncooled microbolometer thermal camera, and a structured-light or time-of-flight LiDAR sensor under controlled conditions. Specimens will span pitted and non-pitted cases to support binary and severity-graded labelling.

All acquisitions will be spatially co-registered to a common reference frame via target-based multi-sensor calibration. Ground-truth annotations will mark pit locations, pit severity categories (shallow, moderate, severe), and corrosion-free background regions, produced through a combination of automated thresholding and manual expert labelling.

C. Fusion Architecture

The proposed framework adopts a **hybrid fusion strategy**, integrating feature-level cross-modal fusion with a late-stage decision refinement mechanism, as illustrated in Fig. 1. Three parallel encoder branches process the respective modalities:

- **RGB branch:** A pre-trained CNN backbone (e.g., ResNet-50 or EfficientNet-B4) extracts hierarchical visual and texture features.
- **Thermal branch:** An analogous CNN backbone processes single-channel thermal images, extracting thermal gradient and anomaly maps.

- **LiDAR branch:** A PointNet++ or sparse 3D CNN encoder processes voxelised or range-projected LiDAR data, extracting surface curvature and depth deviation features.

Intermediate feature maps from the three branches are concatenated at a fusion layer and passed through a shared decoder (Feature Pyramid Network or U-Net-style architecture) for pixel-level corrosion segmentation. A cross-modal attention module will be investigated to learn adaptive, context-dependent fusion weights, enabling the model to dynamically prioritise modalities based on local signal quality.

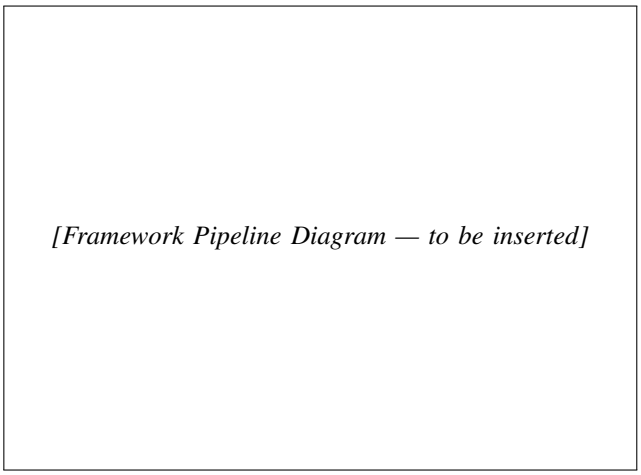


Fig. 1. Proposed trimodal fusion pipeline for pitting corrosion detection. Parallel CNN/PointNet++ encoders process RGB, thermal infrared, and LiDAR inputs. Features are fused via cross-modal attention and decoded to produce a pixel-level corrosion segmentation map with pit severity classification.

D. Evaluation Plan

Model performance will be assessed using the following metrics:

- *Pixel-level:* Intersection over Union (IoU), Dice coefficient.
- *Detection-level:* Precision, Recall, F1-score.
- *Localisation:* Mean Average Precision (mAP) at IoU thresholds of 0.50 and 0.75.

Ablation experiments will systematically remove one or two modalities to isolate each sensor's contribution to overall performance. Robustness evaluation will introduce controlled degradations—simulated sensor noise, illumination variation, partial surface occlusion—to assess performance under realistic offshore conditions. Statistical significance of performance differences between models will be assessed using paired t-tests or Wilcoxon signed-rank tests across cross-validation folds.

IV. Expected Outcomes and Significance

This research is expected to deliver the following contributions:

1) **Novel trimodal fusion framework:** A first-of-its-kind ML pipeline unifying RGB, thermal, and LiDAR data for offshore pitting corrosion detection, with a modular architecture extensible to other structural inspection domains.
2) **Benchmark experimental comparison:** Quantified performance comparisons between the proposed framework and single- and dual-modality baselines, providing the first empirical evidence for the value of trimodal fusion in this specific application.
3) **Multimodal corrosion dataset:** A curated, co-registered collection of RGB, thermal, and LiDAR observations of corroded steel specimens, intended for public release to address current data scarcity in the field.
4) **Practical deployment guidance:** Analysis of trade-offs between detection performance, sensor hardware cost, and computational overhead, informing feasibility assessments for real-world offshore inspection system design.

Offshore structural inspection currently costs the industry several billion US dollars annually, and undetected corrosion remains a primary cause of structural incidents [22]. An automated, high-accuracy multimodal inspection framework could reduce reliance on costly and hazardous manual inspection, support higher inspection cadence, and provide earlier warning of critical corrosion progression—directly advancing both structural safety and operational efficiency in the offshore oil and gas sector.

V. Research Timeline

The proposed research spans approximately 24 months from May 2026 to April 2028, organised into six sequential phases as summarised in Table I. Phases may overlap where parallel tasks are feasible. Intermediate publication of results is planned during Phase 5, targeting IEEE conferences or journals in the areas of non-destructive testing, structural health monitoring, or computer vision applied to engineering.

TABLE I
Research Timeline (May 2026 – April 2028)

Phase	Period
1. Literature Review & Review Paper	May – Jul 2026
2. Research Proposal	Jul – Sep 2026
3. Programming & Tooling Setup	Sep – Nov 2026
4. Dataset Acquisition & Preprocessing	Nov 2026 – Mar 2027
5. ML Model Development & Evaluation	Mar – Dec 2027
6. Thesis Writing & Dissemination	Jan – Apr 2028

VI. Conclusion

This paper has presented a research proposal for a multimodal ML framework for pitting corrosion detection in offshore infrastructure, integrating RGB, thermal infrared, and LiDAR sensor data through a hybrid feature-level fusion architecture. The proposed work addresses a documented gap in the literature—no prior study has unified all three modalities for

this application—and is expected to deliver a novel, validated detection framework, a multimodal benchmark dataset, and practical guidance for offshore structural health monitoring system design. Successful completion of this research has the potential to meaningfully advance inspection safety, reduce operational costs, and improve the early detection of critical corrosion events in the offshore oil and gas industry.

REFERENCES

[1] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[2] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[3] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[4] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[5] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[6] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[7] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[8] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[9] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[10] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[11] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[12] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[13] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[14] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[15] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[16] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[17] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[18] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[19] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[20] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[21] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.
[22] [PLACEHOLDER] Author(s), "Title," *Journal/Conference*, vol. X, no. Y, pp. ZZ–ZZ, Year.